Exploratory Data Analysis:
Shifting Tornado Hazard in the United States and Emerging Insurance Exposure

Merrimack College
Data Science Capstone
DSE6311OM_SP2026R2

Jillian Baker
April 2026

1 Recap Background and Questions

This project examines whether tornado hazard in the United States is shifting from the traditional Great Plains Tornado Alley toward the Southeast's Dixie Alley region over time, and whether counties predicted to exhibit elevated tornado hazard in the following year also have greater present-day insurance exposure based on population, housing units and median home value.

The hypothesis is that tornado hazard is shifting from the Tornado Alley region toward Dixie Alley due to increasing tornado frequency and severity in the Southeast over time. It is predicted that counties located in Dixie Alley will be more likely classified as high tornado hazard in the following year compared to Tornado Alley and will exhibit a higher present-day insurance exposure.

2 Methods

The historical tornado data was obtained from the National Oceanic and Atmospheric Administration (NOAA) Severe Weather Database file (1950-2024) and downloaded as a CSV file. It includes detailed tornado records across the United States including the event date (year, month and day), location, tornado intensity (F-Scale or EF-Scale) and estimated property loss. County-level exposure data was obtained using the tidycensus package in R and the American Community Survey (ACS) 5-year estimates for the variable 2024 for Estimate Total Population (B01003_001), Estimate Total Housing Units (B25001_001) and Estimate Median Value (Dollars) (B25077_001). The datasets were merged using the common FIPS identifier to combine datasets with historical tornado data and present-day exposure variables.

The tornado dataset was cleaned by creating a new column for the complete five-digit FIPS code, combining the state FIPS (stf) and county FIPS (f1-f4) columns, and restoring any leading

zeroes omitted. The originating county (f1) was retained to simplify analysis at the county-year level. Tornado data was filtered for the years 1979-2024, standardized across the F-Scale and EF-Scale to a common scale of 0-5, and aggregated at the county-year level to create hazard-related features for tornado counts, intensity, property loss and the tornado statistics (average length and width). Other feature engineering included regional classification, normalized tornado frequency and exposure-relevant measures for analysis and modeling.

Exploratory Data Analysis (EDA) was conducted in R and included data cleaning, transformation and engineering of new variables (described in detail in the Data Dictionary in the Appendix), as well as univariate, bivariate and multivariate analysis to understand patterns and relationships in the data prior to predictive modeling.

Bivariate analysis included examining year and tornado count trends over the years (1979-2024). Additionally, a bivariate jitter plot was created to examine distribution between region and normalized tornado frequency per 100,000 population, as well as per 10,000 housing units allowing for the identification of outliers.

Multivariate analysis included examination of regional trends with the number of tornadoes over the years (1979-2024) within Tornado Alley and Dixie Alley, as well as strong tornado counts by year (F/EF3 and greater).

Normalized variables were created to account for differences in population and housing exposure. This allows for better comparison of tornado frequency across regions.

3 Results

Variable	Mean	Median	Standard Deviation	Min	Max	Skewness
population_estimate	140,089	30,986	459,866	33	9,808,667	11.27

housing_unit_estimate	59,799	14,386	183,371	33	3,650,466	10.22
medvalue_estimate	208,956	176,100	111,611	48,600	1,633,900	2.55
tornado_count_county_year	1.67	1.0	1.33	1	43	4.80
average_intensity_county	0.65	0.5	0.76	0	5	1.30
max_intensity_county	0.8	1	0.90	0	5	1.13
total_property_loss_year	553,729	4	19,259,400	0.0001	1,960,040,000	80.17
average_property_loss	329,559	3	12,045,790	0.0001	1,504,000,000	104.41
strong_tornado_count	0.54	0	0.25	0	4	5.17
property_loss_per_tornado	296,794	3	11,925,840	0.0001	1,504,000,000	109.13
avg_length	3.18	1.08	5.69	0.00	124	5.17
avg_width	116.5	50	1.86	0.00	3,330	4.54
tornadoes_per_100k_population	14.42	4.46	55.60	0.010	6,060.61	51.86
tornadoes_per_10k_housing_units	2.76	0.95	7.78	0.0027	606.06	24.81

Table 1: Continuous Variables with mean, median, standard deviation, min, max and skewness

Table 1 details the continuous variables in the county-year tornado hazard dataset. Several variables, particularly property loss measures and normalized tornado rate ratios (tornadoes per 100,000 population and tornadoes per 10,000 housing units) were right-skewed. This indicates that most county-year observations that have a significantly small number of tornadoes or property loss and a smaller amount of county-years have a large amount.

Variable	Category	Count	Percent	Proportion
region_category	Tornado Alley	7,843	24.66%	0.2466

region_category	Dixie Alley	6,395	20.11%	0.2011
region_category	Other	17,566	55.23%	0.5523
high_tornado_hazard	0 = Not High Hazard	23,853	75%	0.75
high_tornado_hazard	1 = High Hazard	7,951	25%	0.25
high_tornado_hazard_next_year	0 = Not High Hazard	21,617	67.97%	0.6797
high_tornado_hazard_next_year	1 = High Hazard	7,177	22.57%	0.2257
high_tornado_hazard_next_year	NA	3,010	9.46%	0.0946

Table 2: Categorical variables with count, percent and proportion

Table 2 shows that approximately 25% of the county-year observations are high tornado hazard for both current and next year observations and approximately 75% were not high hazard. Approximately 22% of county-year observations are indicated as high tornado hazard next year. More than 50% of county-year observations occurred outside of Tornado and Dixie Alley with approximately 25% occurring in Tornado Alley and 20% in Dixie Alley. The NA value for high_tornado_hazard_next_year looking one year into the future for the last observed year in each county, there is no future data available.

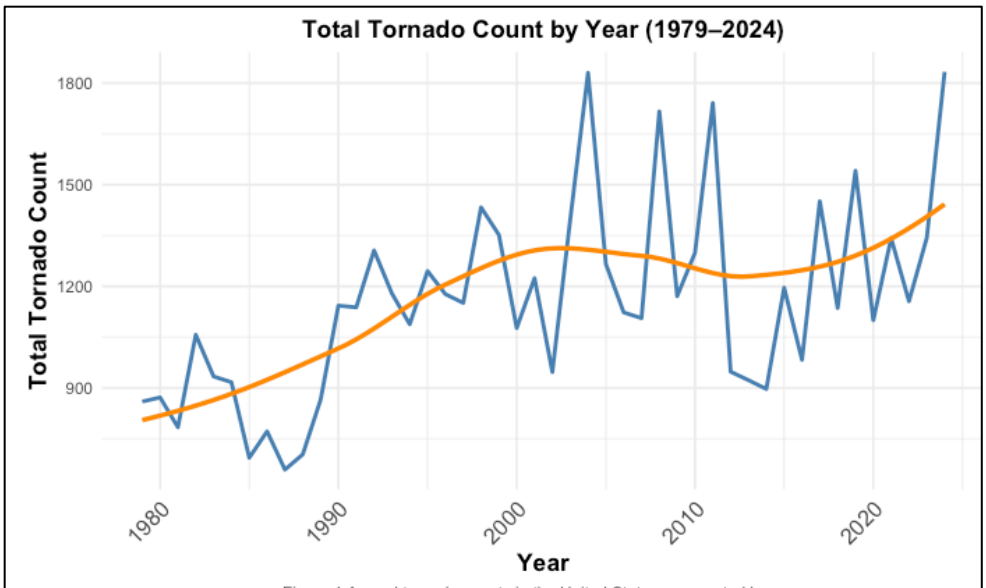

Figure 1 Annual tornado counts in the United States aggregated by year

Figure 1 examines the annual tornado count in the United States between 1979-2024. There is significant variability from year to year with the frequency of tornadoes. This broad overview suggests that tornado frequency appear to have increased in recent years, particularly from 2015 onward. However, variability and fluctuation of tornado frequency over time is expected.

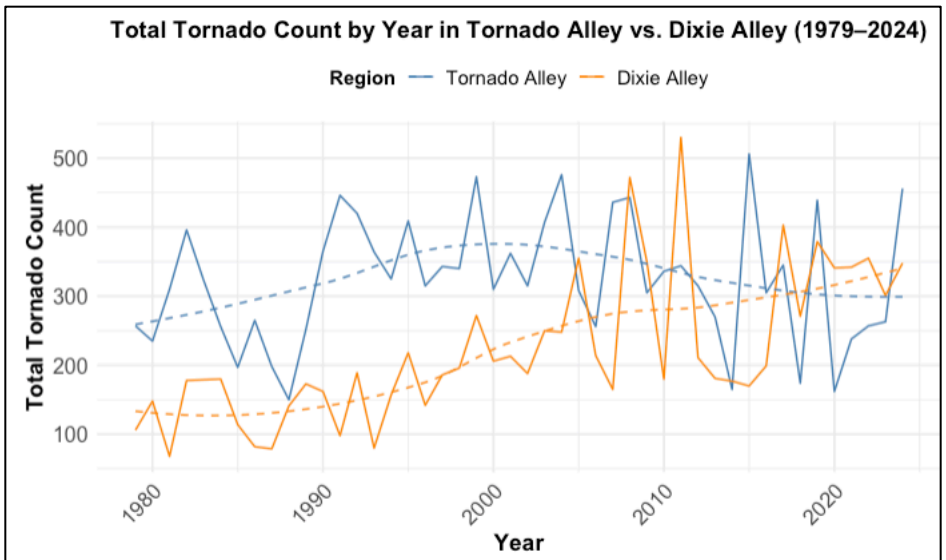

Figure 2 Annual tornado counts by year in Tornado Alley & Dixie Alley

Figure 2 shows the annual tornado counts over time in both Tornado Alley and Dixie Alley between 1979-2024. The trend lines suggest that tornado counts in Tornado Alley may have remained stable or slightly declined in recent years, while tornado counts in Dixie Alley appear to have increased in frequency, particularly from 2000 onward. This pattern is consistent with the project hypothesis that greater tornado hazard events may be shifting toward Dixie Alley over time.

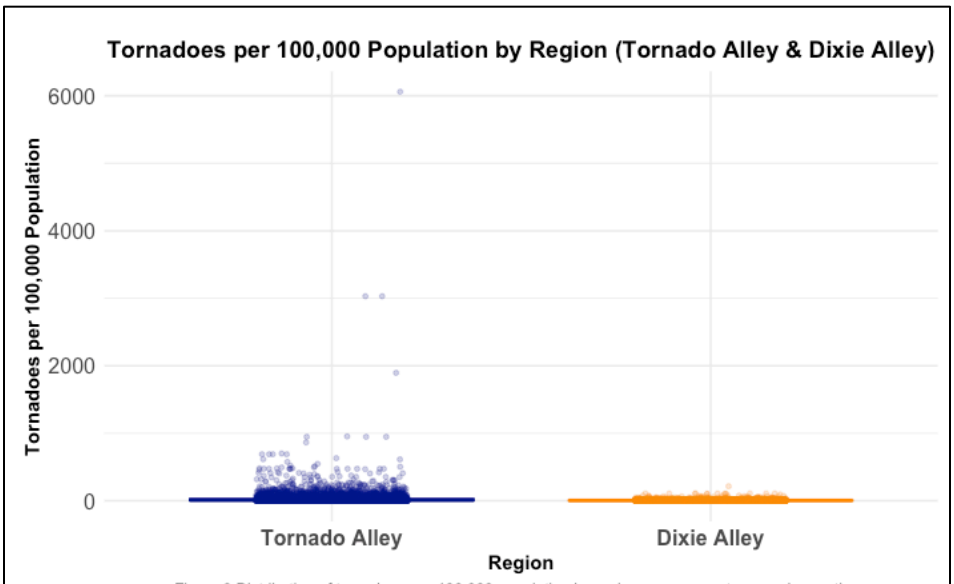

Figure 3 Distribution of tornadoes per 100,000 population by region across county-year observations

Figure 3 shows the distribution of normalized tornado frequency per 100,000 population across county-year observations in Tornado Alley and Dixie Alley. There is extreme right-skewing within both regions, indicating that most county-years experience low normalized tornado frequency (as detailed with more values clustering near the bottom of the plot), in addition to a few extreme outliers. Outliers occur more often in Torndo Alley which indicates tornado occurrences within counties of low population. Dixie Alley appears to be more consistently concentrated.

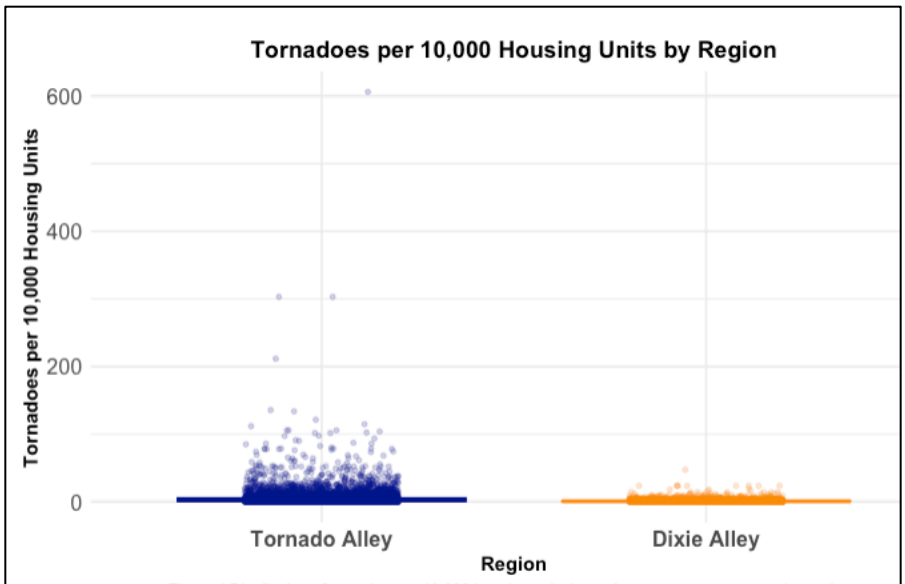

Figure 4 Distribution of tornades per 10,000 housing units by region across county-year obsevations

Figure 4 shows the distribution of tornadoes per 10,000 housing units across county-year observations in Tornado Alley and Dixie Alley. Similar to Figure 3, there is also highly right-skewed within both regions, indicating that most county-years experience low normalized tornado frequency (as detailed with more values clustering near the bottom of the plot), in addition to a few extreme outliers. Outliers also occur more often in Tornado Alley which indicates tornado occurrences within counties with lower housing units. Dixie Alley appears to be more consistently concentrated.

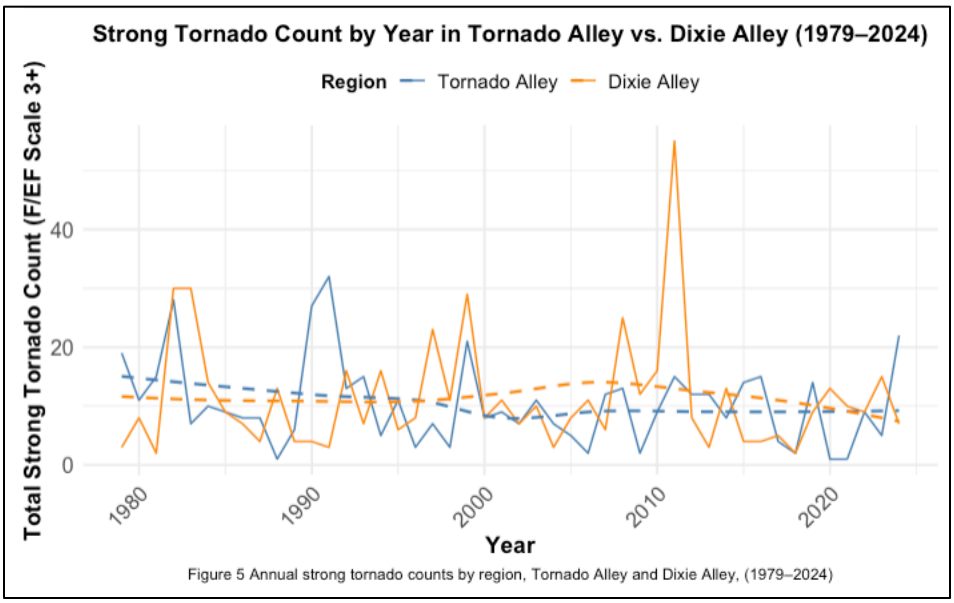

Figure 5 Annual strong tornado counts by region, Tornado Alley and Dixie Alley, (1979–2024)

Figure 5 shows the annual strong tornado counts (F/EF3 and greater) in Tornado Alley and Dixie Alley between 1979-2024. Both regions exhibit a substantial varaibiltiy of strong tornado frequency over the years. Tornado Alley appears somewhat more elevated than Dixie Alley in the earlier years, has fewer spikes and appears to trend stable or slightly decreased over the years. In comparison, Dixie Alley shows notable spikes in strong tornadoes and has trended slightly higher than Tornado Alley occurrences, especially in more recent years. It appears they have similar occurrences of strong tornadoes most recently. This relationship suggests that strong tornado activity may be more distributed over other regions over time.

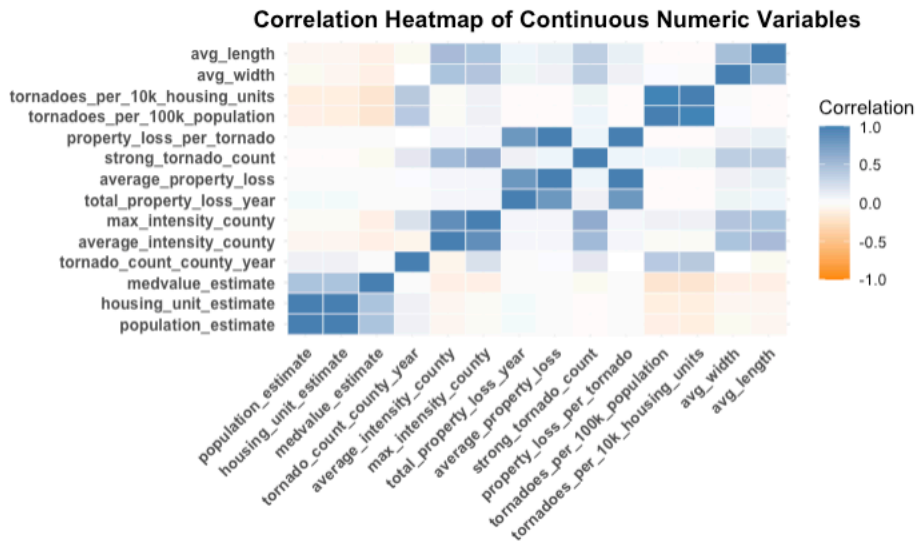

Figure 6 A correlation matrix of continuous numeric variables included in the analysis

Figure 6 details a correlation matrix of the continuous numeric variables that are included in the project analysis. There are strong positive relationships observed, especially between population estimate and housing unit estimate, average and maximum tornado intensity and the normalized tornado rate measures (tornadoes per 100,000 population and per 10,000 housing units. Some variables may overlap and will need to be considered during modeling for possible multicollinearity.

4 Discussion and Next Steps

The exploratory analysis performed provides preliminary support to the research question and hypothesis that tornado hazards in the United States may be gradually shifting towards the Southeast's Dixie Alley region over time. Figure 2 suggests that the annual tornado counts in Tornado Alley may have declined in the more recent years, while tornado counts within Dixie Alley has been gradually increasing. Figure 5 suggests that stronger tornadoes (F/EF3 and greater) have been recorded more frequently within the Dixie Alley region, specifically during the more recent years, although both regions show considerable variability over time. Tornado

Alley also exhibits greater outliers with the exposure data for population and housing units. In Figure 6, the correlation matrix shows strong positive relationships between housing units and population estimates as well as the normalized measures for tornadoes per 10,000 housing units and 100,000 population.

The next steps for this project will include data cleaning and preprocessing for predictive modeling. Variables will be reviewed for multcollinearity and preparing the binary target variable for predicting next-year high tornado hazard classification. The target variable is high_tornado_hazard_next_year, which is a binary classification (1 = high tornado hazard, 0 = not a high tornado hazard). This will indicate whether a county will be classified as an elevated tornado hazard in the following year. This was created based on a composite hazard score derived frm standardized county-year tornado frequency, intensity, strong tornado count and property loss measures. Any county-years at or above 75% of the composite score are indicated as high hazard and this was shifted forward one year for each county for the predictive target variable.

The EDA has shown there are many highly right-skewed variables, these will be reviewed for appropriate transformations. Predictive modeling will include, logistic regression as the starting model in addition to random forest and XGBoost as flexible models to analyze if performance improves when compared to the logistic regression model. The models will be evaluated using recall and PR-AUC. Recall is prioritized as the primary evaluation metric to ensure identification of high-risk counties. Confusion matrices will provide a visualization of model performance. The data will be split using county-year observations, the training set will include the earlier years of 1979-2014 and the testing set will use the later years of 2015-2024. This will help ensure the model is evaluated on future data. However, it will also consider

subgroup-aware cross-validation. This will ensure that when cross-validation is performed that it is subgroup-aware and observations for Tornado Alley and Dixie Alley are included. This will be perfomed with a stratified k-fold rather than a conventional k-fold. Hyperparameter tuning will be conducted using cross-validation (randomized search) to optimize the model performance and reduce the risk of overfitting.

As the data contains strong geographic structure, the model may overfit by learning spatial dependence across counties rather than relying on the intended variables and exposure variables. This will not be fully addressed in this capstone, however, future work could include incorporate spatial lagging of predictors to better account for geographic relationships.

5 GitHub

https://github.com/jill-baker/tornado-insurance-risk-repo

6 Appendix

The Data Dictionary below includes identification of key raw, engineered and target variables in the final county-year tornado hazard and insurance exposure dataset. The target variable, high_tornado_hazard_next_year, was created from the hazard-related variables in the dataset: tornado_count_county_year, average_intensity_county, max_intensity_county, total_property_loss_year and strong_tornado_count.

Column Name	Example	Data Type	Description
om	'192'	Integer	Tornado number. Count of tornadoes during given year from NOAA dataset
yr	'1950'	Integer	Year of tornado event (1950-2024)
mo	'3'	Integer	Month of tornado event, (1-12)
dy	'26'	Integer	Day of tornado event, (1-31)

date	'1950-03-26'	Date	Date of tornado event in yyy-mm-dd format
st	'AR'	String	State (two-letter postal abbreviation) of tornado event
mag	'2'	Integer	F-Scale (EF-Scale after 01/07) Values 0, 1, 2, 3, 4, 5, -9 (-9=unknown)
loss	'4'	Numeric	Estimated property loss in millions of dollars. 0 or blank = unknown, not $0
len	'17.4'	Numeric	Length of tornado in miles
wid	'150'	Integer	Width of tornado in yards
county_fips	'05019'	String	Federal Information Processing Standards code for county, full 5 digit code (first 2 digits for state, last 3 digits for county).
county_name_state	'Clark County, Arkansas'	String	Full county name and state
population_estimate	'21125'	Integer	**Predictor.** Present-day exposure variable. Estimated total population in county
housing_unit_estimate	'10029'	Integer	**Predictor.** Present-day exposure variable. Estimated total number of housing units in the county
medvalue_estimate	'155000'	Numeric	**Predictor.** Present-day exposure variable. Estimated median home value in the county (in dollars)
region_category	'T'	String	**Predictor.** Indicates if county is located in Dixie Alley, Tornado Alley or Other. Dixie Alley = D, Tornado Alley = T, Other = O.
high_tornado_hazard_next_year	'1'	Binary	**Target variable.** Indicates whether a county is classified as high tornado hazard in the following year based on county's forward-shifted composite hazard classification (1=high tornado hazard, 0=not high tornado hazard)
tornado_count_county_year	'4'	Integer	*Target creation hazard variable.* Total number of tornadoes per county for a given year
average_intensity_county	'3'	Numeric	*Target creation hazard variable.* Average tornado intensity (F/EF-Scale) for tornadoes in county for a given year.
max_intensity_county	'5'	Integer	*Target creation hazard variable.* The maximum/highest observed tornado intensity per county in given year
total_property_loss_year	'5'	Numeric	*Target creation hazard variable.* The total property loss with all occurred tornadoes per county in given year

average_property_loss	'4'	Numeric	Hazard variable. The average property loss for tornadoes in county for a given year
strong_tornado_count	'3'	Integer	*Target creation hazard variable.* Count of tornadoes in county-year where the tornado intensity (mag) is >=3
property_loss_per_tornado	'50000'	Numeric	Calculation of total_property_loss_year / tornado_count_county_year
tornadoes_per_100k_population	'100'	Numeric	Normalized measure: Calculation of tornado_count_county_year / population_estimate * 100000
tornadoes_per_10k_housing_units	'50'	Numeric	Normalized measure: Calculation of tornado_count_county_year / housing_unit_estimate * 10000
avg_length	'5'	Numeric	**Predictor.** Average length in miles for tornadoes per county
avg_width	'5'	Numeric	**Predictor.** Average width in yards for tornadoes per county
composite_hazard_score	' -0.50'	Numeric	Composite index created by averaging standardized tornado frequency, intensity, and loss variables. Used to classify high hazard counties.
high_tornado_hazard	'1'	Binary	Variable used to create the predictive target high_tornado_hazard_next_year and represents the current-year classification

7 References & Data Sets

- National Oceanic and Atmospheric Administration (NOAA), National Weather Service. (2024). *Severe Weather Database Files (1950–2024)*. Storm Prediction Center. https://www.spc.noaa.gov/wcm/
- U.S. Census Bureau. (2024). American Community Survey 5-Year Estimates. https://www.census.gov/programs-surveys/acs
  - Tidycensus in R with the following variables selected at the county-level: Estimate Total Population (B01003_001), Estimate Total Housing Units (B25001_001) and Estimate Median Value (Dollars) (B25077_001) for the year 2024
- Gagan, J. P., Gerard, A., & Gordon, J. (2010, December). *A HISTORICAL AND STATISTICAL COMPARISON OF "TORNADO ALLEY" TO "DIXIE*

*ALLEY."*https://nwafiles.nwas.org/file/nwafiles/digest/papers/2010/Vol34No2/Pg145-Gagan etal.pdf. http://nwafiles.nwas.org/digest/papers/2010/Vol34No2/Pg145-Gagan-etal.pdf

- US Department of Commerce, N. (2018, October 17). *Is tornado frequency increasing in parts of the U.S.?*. National Weather Service. https://www.weather.gov/lmk/niu_tornado_frequency_study
- Nolan, D. (2024, March 29). *'Dixie Alley': Tornadoes increasing in Tennessee, study finds*. https://www.wkrn.com/weather-headlines/dixie-alley-tornadoes-tennessee/.
- US Department of Commerce, N. (2018a, August 14). *Enhanced Fujita scale*. National Weather Service. https://www.weather.gov/tae/ef_scale